

Final Project

William Pascoe

CYB-250: Cyber Defense

Aries Preston

April 19, 2026

Final Project

Personnel or Human Factor Trend

When it comes to cybersecurity and keeping networks safe the human element has always been the weakest link. As a result of this companies have moved toward a trend of more continuous bite size training as opposed to a once-a-year daylong seminar. This makes the training easier to follow and by reinforcing it throughout the year, and employees are more likely to retain the information. The continuous training helps the employees be on the watch for situations like phishing emails shoulder surfing or any other type of attack helps to keep the company data safe.

Attackers could use techniques like phishing, or shoulder surfing to attack the data that is being transmitted on the headset device. Continuous education and awareness training should include appealing to a sense of urgency. Phishing attempts often include wording meant to entice the reader to click on a link (Sherr & Sherr, 2025). This includes items such as saying a package cannot be delivered, or there is outstanding payment due. Employees need to be trained in how to mitigate this. They should also be trained in how to be more aware of their surroundings and try to find out if someone is trying to look at what they are seeing. As a result of this training, they can recognize phishing or shoulder surfing types of attacks and thereby keep the company's network and data secure.

This trend of continuous bite size training has certain risks and rewards associated with it. Some of the rewards of this training are seeing a reduction in successful phishing scams. Part of the training should include practical simulations using test phishing emails from the security department. As more people learn the technique of phishing emails and how to properly handle them, departments should have a decrease in people clicking the links or opening the malicious

attachments. A risk also associated with this type of training is fatigue. People could constantly feel like they are being tested and monitored and that could cause them to quickly grow tired of a new routine.

The emerging trend of training is shifting from annual compliance-based training to real-time, event-driven education. This impacts policies by moving away from punitive measures toward positive reinforcement, and it forces technologies to integrate better with learning management systems.

Data Protection Strategy or Technology

The data protection strategy that can help provide security to an organization is a zero-trust philosophy (Terry, 2025). This type of security framework that mandates strict network, verification for every user and device attempting to access a network resource, regardless of whether they are inside or outside the organization's network. The National Institute of Standards and Technology developed the standard NIST 800-207 as a guide to implement this strategy.

The core principles of the zero-trust model are to continually verify, limit the blast radius, and automate collection and response (Terry, 2025). With continuous verification you should verify whether it is a human, device, or application. None of these are to be trusted by default. Limiting the blast radius means that if there is an event breach it is as contained as possible. When the standard talks about automate collection and response it emphasizes the automation of context collection and real-time response to ensure that the security system can react swiftly and accurately to potential threats.

With the zero-trust strategy the data remains encrypted and any data that is stolen is unusable because it will still be ciphertext until it reaches the final endpoint of the device itself.

This makes a man in the middle attack against the Bluetooth transmission ineffective for any data that is intercepted. Users should understand not to transmit anything over any public Wi-Fi connections though to make sure that they are secure as possible.

One of the major rewards of the zero-trust model being able to eliminate the trusted part of the network with the untrusted part of the network. If we assume that all the network is untrusted it allows us to better prepare the network for intrusion. If an attacker were able to get into a trusted side, they might have an easier time getting data because we could assume that part of the network does not need to be as protected. In our scenario we assume the whole network needs to be secured and we do not have things set up like DMZ's or untrusted zones. The risk of having this type of setup is added to the complexity. If we add more firewalls, tighter switches, and router security practices it can cause more work to set up and get newer equipment online and functioning properly. Also having more security means more users can be restricted from doing their jobs by not getting access to essential systems they need or even the possibility of getting locked out of a system more frequently by forgetting the longer passphrases they need to access the system.

The cryptographic technique used with a zero-trust model is encryption on all levels of the network. Whether the data is at rest or in transit the data should be encrypted to the highest standard possible. When accessing the system users should use authentication that requires use of a public / private key asymmetric login and even consider using multifactor authentication to verify the user's identity.

The advantage of using encryption at all levels is that we ensure that the integrity and confidentiality of the data remain intact. If the data is stolen at some point the data would be in cyphertext and not much use to an attacker. However, a disadvantage to this process would be

more resources could be required. When we transmit the data, we must ensure that it is properly encrypted then when it reaches its destination it must then be decrypted to be useable for the user requesting it. This requires more processing power and users may experience some latency in retrieving the data while this process occurs. Also, when we use things like private keys, we must ensure that those keys are stored in a manner that they cannot be stolen and used to decrypt any data an attacker may acquire.

One of the security concerns with using encryption is the possibility of using outdated protocols. If a company is not keeping up with updating their protocols, they could be subject to new vulnerabilities that are constantly being discovered. As better computers, or quantum computers, are placed online security protocols must become more advanced because the algorithms that are currently being used can easily be broken. Another concern is if an attacker can get into an endpoint and subsequently access the security tunnel, they could install a backdoor that intrusion detection might not pick up because its installation hides in the symmetric data transfer.

System Protection

Using endpoint and server protection technologies is a viable way to protect the system. Endpoint detection and response, also referred to as endpoint detection and threat response (EDTR), is an endpoint security solution that continuously monitors end-user devices to detect and respond to cyber threats (Cross 2024). Using a SIEM as a partner to endpoint technology helps keep the server side of the data secure by alerting them to any intrusions that may occur in the system. Both security applications focus on incident detection and response.

Having endpoint detection either in the phone app or on the device itself would be an effective way to ensure that the data transmitted from the server has its integrity intact. If there is

any malicious activity during the transmission of data, the EDR would alert the user of tampering. The SIEM would alert staff if there is an attempt to access or manipulate data on the server side.

Some of the risks and rewards that are associated with EDR's and SIEM's are the ability to roll devices and alert fatigue. Having the ability to look at endpoints and roll back to a known safe point is a great asset when dealing with an EDR. The other side of the scenario is that this kind of set up could lead to a lot more false positives which can create alert fatigue for SOC analysts. Too many alarms can make them complacent when a genuine true positive comes along.

The emergence of XDR, or Extended Detection and Response is unifying endpoint, network, and cloud data into a single stream. This technology is forcing a shift in security strategies from a strategy of manual alert correlation to automated, unified detection in a centralized Security Operations Center, or SOC (*Understanding the Difference between EDR, SIEM, SOAR, and XDR*, 2024).

References:

- Cross, K. (2024, September 10). *SIEM vs EDR: Similarities and Differences* | CrowdStrike. CrowdStrike.com. <https://www.crowdstrike.com/en-us/cybersecurity-101/next-gen-siem/siem-vs-edr/>
- Terry, R. (2025, March 13). *What is Zero Trust Security? Principles of the Zero Trust Model* | CrowdStrike. CrowdStrike.com. <https://www.crowdstrike.com/en-us/cybersecurity-101/zero-trust-security/>
- Scherr, D., Scherr, T. M., & Scherr and Tanya, D. (2025, March 5). *The Human Factor in Cybersecurity Events: Critical Education Components - Domestic Preparedness*. Domestic Preparedness. <https://www.domesticpreparedness.com/articles/the-human-factor-in-cybersecurity-events-critical-education-components/>
- Understanding the Difference Between EDR, SIEM, SOAR, and XDR. (2024, August 9). Sentinel One. <https://www.sentinelone.com/cybersecurity-101/xdr/understanding-the-difference-between-edr-siem-soar-and-xdr/>